AREI MOHAMMED

Software Engineer | AI Systems & LLM Development

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SUMMARY

Software Engineer building government digital services at national scale, currently working on HMRC's New Computerised Transit System in Scala and the Play Framework. Experienced across backend APIs, microservices and accessible front ends, with Python used for tooling and data pipelines, including a parser and generator now powering the live HMRC developer portal. Works test-first, supports production systems directly (75+ automation robots in HMRC), and enjoys explaining technical work to non-technical stakeholders. Builds AI applications in my own time using the Anthropic Claude API, Redis and PostgreSQL. Designs and teaches a TDD curriculum for junior developers. MSc Computer Science (UCL), BEng Aerospace Engineering (City, University of London).

EDUCATION & CERTIFICATION

UCL, University College London September 2022 – September 2023

MSc, Computer Science | Databases · Functional Programming · Algorithms · Computer Architecture · Software Engineering

City, University of London September 2019 – May 2022

BEng, Aerospace Engineering | Mathematics · Aerodynamics · Structural Mechanics · Control Systems · Machine Learning

KEY SKILLS

- Python · React · Next.js · Play Framework
- Reinforcement Learning · PPO · Policy Gradient
- Scala · Java · Spring Boot · Node.js · TypeScript
- Infrastructure · PostgreSQL · Redis · Docker
- AI Systems & Automation · NLP integration
- Anthropic Claude API · LLM tool-use · RAG
- SQL · JavaScript · Python · C · MATLAB
- Agile · Scrum · JIRA · estimation & planning
- End-to-end service ownership
- Prompt engineering · Langfuse eval
- MATLAB · Python RL frameworks · CFD integration
- Development Practices · TDD · JUnit 5
- SageMaker · IBM Cloud · CI/CD · GitHub Actions · Jenkins
- Conversational AI · RPA (NiCE) · Workflow automation
- Agile/Scrum · Technical mentoring · Curriculum design
- Numerical modelling · Simulation
- Numerical modelling · MATLAB · simulation
- Technical & business stakeholder management

PROFESSIONAL EXPERIENCE

HMRC February 2026 – Present

Automation Engineer / RPA Developer | London, UK

- Built and maintained complex automation robots across high-volume HMRC business areas including Self-Assessment, Fraud and Pensions, using NiCE Automation Studio to automate case processing on critical tax administration workflows.
- Reduced technical failure rate of the most important HMRC automation machine from 16% to 9%.
- Monitored, triaged and supported over 75 production robots, diagnosing failures, applying fixes and ensuring continuity of automated services processing taxpayer data at national scale.
- Maintained and administered the SQL databases underpinning the automation estate, tuning queries and safeguarding data integrity across the platform.
- Produced operational reporting and analytics using IBM Cognos to track robot performance, throughput and failure trends for senior stakeholders.

Mercator Digital July 2024 – Present

Software Engineer | London, UK

- Developed and maintained critical national infrastructure APIs in Scala using the Play Framework within the HMRC New Computerised Transit System (NCTS) team, covering Common Transit Convention services, transit movement validation, conversion and routing.
- Led and contributed to the Scala 3 migration of multiple HMRC micro services, ensuring backward compatibility and reducing technical debt across government-scale codebases.

- Owned deprecation of API v2.0 references across transit-movements-auditing, removing over 6,000 lines of legacy code and significantly improving long-term maintainability.
- Built a Python documentation pipeline for the NCTS Phase 6 Technical Interface Specification, implementing a dangling-else resolution algorithm with lookahead to correctly parse and indent complex IF/THEN/ELSE business rules, generating HTML partials and Ruby ERB templates consumed by the live HMRC developer portal.
- Contributed to the Pension Scheme Return and CDS Reimbursement Claim front ends, delivering accessibility fixes, Welsh language localisation and new user journeys across multiple production releases.
- Mentored and onboarded new engineers and designed and delivered a structured Java training curriculum for junior developer cohorts covering TDD, Spring Boot and HMRC service standards.

IBM July 2023 – September 2023

Software Engineer — Internship | London, UK

- Implemented speech recognition within IBM Watson on a React grocery application as part of the industry exchange programme, improving accessibility for individuals with dysarthric speech patterns.
- Used IBM Cloud services to train and deploy Watson actions, enabling users to interact with the front-end application entirely through voice commands.
- Built the React application from scratch alongside a supporting API handling product availability, collaborating via Git and presenting progress to senior stakeholders.

City, University of London June 2021 – September 2021

Software Engineer — Internship | London, UK

- Worked with engineering professors and PhD students to develop analysis tooling for turbojet and energy research, constructing a Python-based thermodynamic Brayton cycle analysis tool that improved cycle efficiency modelling and accuracy.
- Created a geographical data representation tool consuming Excel and JSON sources to visualise energy consumption across densely populated countries, reducing research time by an estimated 40 hours.

PROJECTS

Deep Reinforcement Learning for Active Flow Control · *MATLAB · PPO · Actor-Critic Networks · Navier-Stokes CFD*

- Designed and implemented a deep reinforcement learning control system that interfaced a Proximal Policy Optimization (PPO) agent with a Navier-Stokes CFD solver, performing active flow control on a 2D cylinder via hinged flaps to minimise the drag coefficient.
- Extended the existing single-flap solver to a two-flap configuration and built the full environment interface — continuous observation and action spaces, custom step and reset functions, and a reward function penalising physically invalid flap overlap.
- Constructed the actor-critic architecture (512-unit networks; continuous stochastic actor with tanh mean and softplus standard-deviation paths) and tuned PPO hyperparameters including clip factor, GAE factor, entropy loss weight and discount factor.
- Trained across 10,000 episodes at laminar and weakly turbulent Reynolds numbers, benchmarking the learnt policy against fixed-angle controls: converged to within 0.003 of the best fixed-angle drag coefficient under laminar flow, while identifying failure to converge under turbulent conditions and a constraint-handling limitation invalidating a significant share of episodes.

Trading Thesis AI · *TypeScript · Node.js · Next.js · PostgreSQL · Redis · Anthropic Claude API*

- Architected and built a full-stack AI system that monitors financial news continuously and evaluates each event against user-defined trading theses, returning structured impact assessments (direction, confidence score, suggested action, risk factors) via the Anthropic Claude API with guaranteed-parseable tool-use output.
- Engineered a multi-source ingestion pipeline (Polygon.io REST polling, Finnhub WebSocket with auto-reconnect, RSS feeds) normalised into a common signal shape, with SHA-256 Redis deduplication and ticker/text relevance matching to minimise redundant LLM calls.
- Implemented asynchronous evaluation via a BullMQ/Redis job queue with controlled concurrency and exponential-backoff retries, dispatching threshold-based email alerts when confidence exceeds a per-thesis bar.
- Shipped multi-tenant authentication with Clerk (user-scoped data, ownership checks), rolling thesis health scores, and a live signal feed with filterable evaluation history in a pnpm/Turborepo monorepo.